BHASHAWRITE

Hinglish Smart Writing Assistant

Roman script → Native script → Clearer writing → Optional English

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CS3235 • Mini Project 4 • CIE3 Lab

Individual submission | Student: [Your name] | USN: [Your USN]

The problem: Roman-script Indian languages

People type informally

“mujhe paani chahiye”
“nanna hesaru”
“enakku thanni venum”

Naive tools fail

Variable Roman spellings
English technical words
Ambiguous proper names
Mixed punctuation

Our solution

Language-aware conversion
Clear model provenance
Measurable evaluation
Usable Streamlit frontend

Design principle: separate transliteration from translation and rewriting.

What the application does

1	**Input**	Roman Hindi, Kannada or Tamil
2	**Convert**	IndicXlit or explicit offline demo
3	**Polish**	Whitespaces and punctuation
4	**Extend**	Optional BharatGen / IndicTrans2
5	**Evaluate**	CER, word accuracy, exact match

Modular, inspectable architecture

INPUT

Streamlit editor

PROTECT

URL • email • numbers

CONVERT

IndicXlit / demo

POLISH

Local correction

OPTIONAL

Param1 + IndicTrans2

OUTPUT

Results + export

Model roles, hardware and honesty

AI4Bharat IndicXlit

Real neural Roman → native transliteration. Optional install and weights. Word-level inference.

BharatGen Param1

Param-1-2.9B-Instruct optional Hindi-only rewriting via a separate vLLM endpoint.

AI4Bharat IndicTrans2

Optional native → English using distilled 200M model and IndicTransToolkit.

Pretrained model inference was NOT benchmarked here; demo baseline ≠ neural accuracy.

Streamlit app: three simple tabs

Writer

Select language and backend
Enter Roman text
Convert and download
See unknown-word warnings

Evaluation

45 reference examples
Run a backend on all rows
CER and word accuracy
Download detailed CSV

Project info

Architecture and models
Which features are optional
Limitations and privacy
Reproducibility notes

Test corpus and methodology

Hindi

15 cases

Kannada

15 cases

Tamil

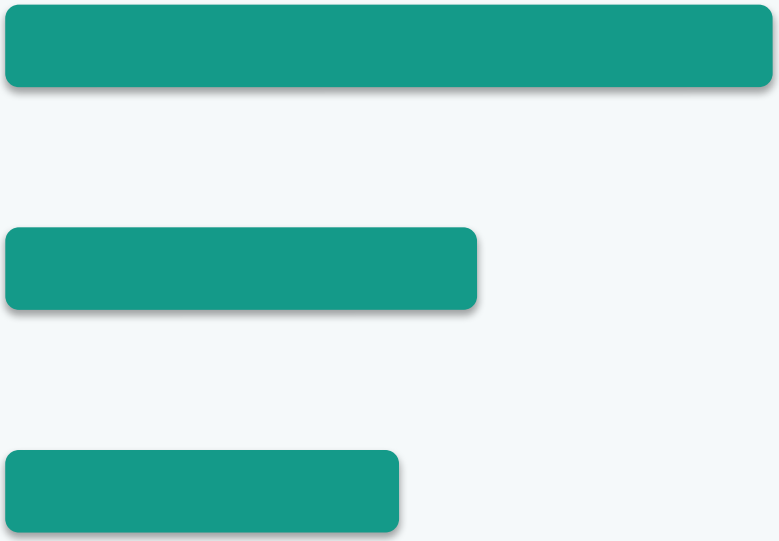
15 cases

Categories: greeting • self-introduction • education • question • request • numeric time • code mixing • unseen words

Tiny, manually authored evaluation set — not the official Aksharantar benchmark.

Measured results: OFFLINE demo only

Mean CER ↓



Hindi	**0.0571**	Words 95.6% · Exact 86.7%
Kannada	**0.0351**	Words 97.8% · Exact 93.3%
Tamil	**0.0293**	Words 97.8% · Exact 93.3%

Overall: CER 0.0405 | Word accuracy 97.04% | Exact match 91.11%

Error analysis: four observed failures

H13	“certificate” retained in Roman script
H14	“Email” preserved as a technical term
K13	“assignment” missing from Kannada lexicon
T13	“assignment” missing from Tamil lexicon

Cause: tiny lexicon and code-mix tradeoffs — not a result of neural inference.

Safety, limitations and future work

Known limitations

Roman spelling ambiguity
Small authored dataset
No official corpus benchmark
Unverified neural setup

Safety controls

Input length limit
Model provenance shown
Untrusted prompt boundaries
No silent fake translation

Next steps

Real IndicXlit benchmark
Native-speaker review
Top-k choice editor
Rate limits and deployment

Reproduce the demonstration

```
python -m venv .venv
pip install -r requirements.txt
streamlit run app.py
python -m pytest -q
python evaluation/run_eval.py --mode demo
```

20 local tests passed • 45 offline-demo cases • source and docs included

What is included and what remains

Delivered & verified

Streamlit source + CLI
45-case CSV & metrics
20 passing pytest tests
PDF report + editable slides
Architecture and failure analysis

Student completion actions

Install / verify real AI4Bharat models
Add actual name and USN
Publish your own GitHub repository
Capture true UI demonstration
Add real model benchmark results

Repository-ready deliverable — no claim of a published GitHub URL.

THANK YOU

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Questions: IndicXlit • transliteration vs translation • CER • error cases • model limits